

Introduction

Earth's rotation causes sunlight to shift across the day, leaving many areas only partially illuminated even during daylight hours. This limits efficiency for applications such as solar panels or plant growth. This project aims to improve sunlight usage by redirecting light from areas with consistent light exposure to shaded or low-light regions, increasing overall solar availability.

Novelty

Our heliostat adds height adjustment as an additional degree of freedom compared to traditional designs. As shown in Fig. 1, the system uses a modular architecture that allows components to be easily modified or replaced while maintaining a compact and lightweight form factor. A convex mirror is also being explored to improve fire safety, though a flat mirror may be used if the reflected light intensity is not found to be hazardous.



Fig. 1: Heliostat Frame's Modular Parts

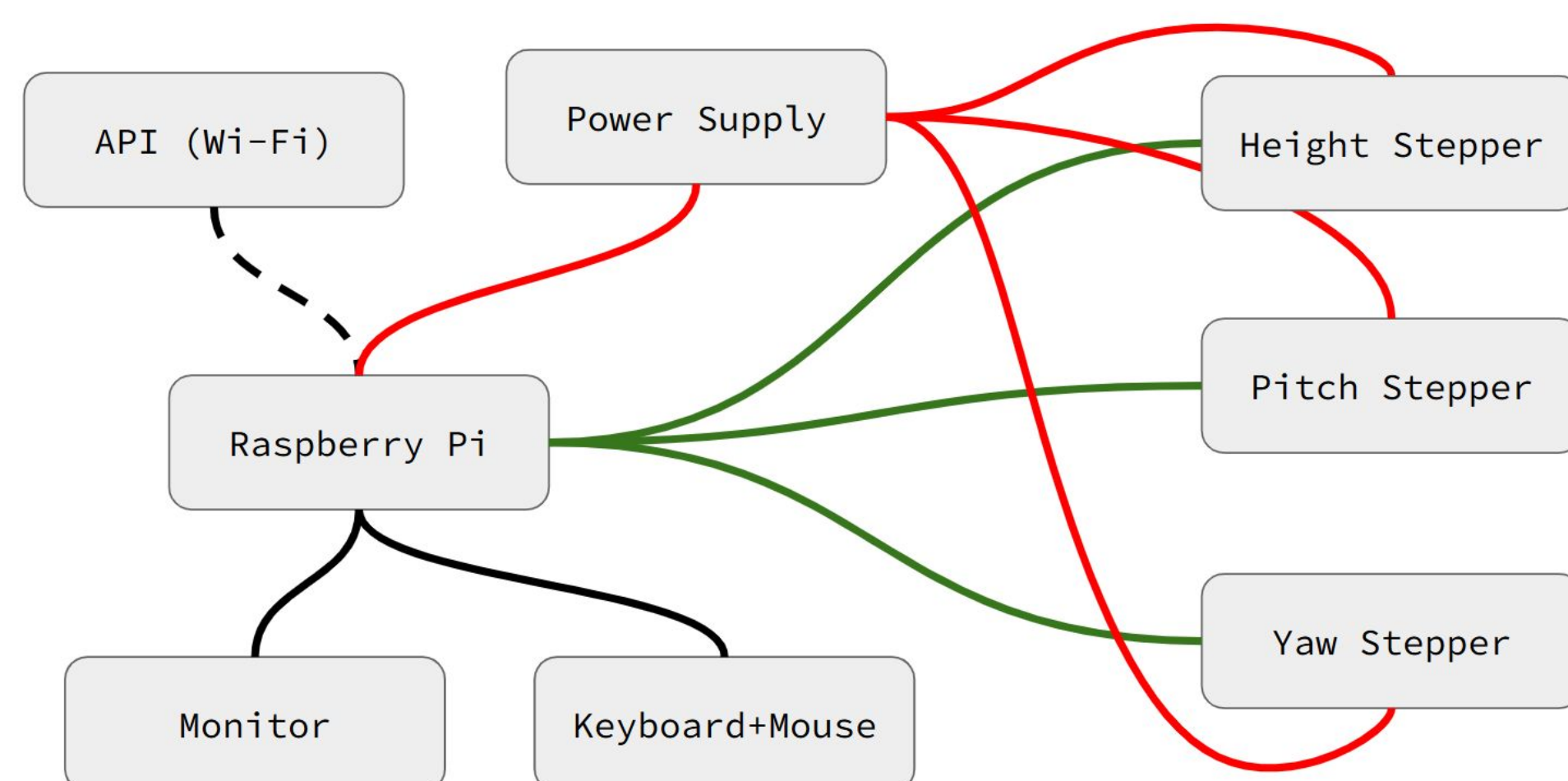


Fig. 2: Physical System Wiring and Wifi-Comms

Micro-Heliostat

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Methods

The overall physical system architecture of the heliostat is shown in Fig. 2, illustrating the integration of control electronics, motor systems, actuator components, the reflective assembly, and user interface used to redirect sunlight. The fully assembled heliostat prototype demonstrates the lightweight structural frame and mirror support system designed to allow rotational movement and vertical adjustment for improved sunlight targeting accuracy. This modular design allowed for quick part replacements as we improved our design. The completed heliostat is in Fig. 3 to the right.

A linear actuator mechanism was incorporated to enable controlled height adjustment of the heliostat, allowing the mirror's pivot position to adapt to changing sunlight angles throughout the day. The system uses stepper motors to achieve precise rotational movement, and further increases its own precision and torque by making use of cycloidal drive gears. The cycloidal drive systems here are used for pitch and yaw, and they give us 11 times the steps and 11 times the torque. The heliostat is controlled by a Raspberry Pi, which handles motor movement and user input. This interface for user input is shown in Fig. 4, and it allows users to monitor and adjust the heliostat's positioning in real time. With this combination of API data from the GeoLocation API, cycloidal drives, and stepper motors, the heliostat is able to angle the mirror to the precise position for sunlight to reach the target.

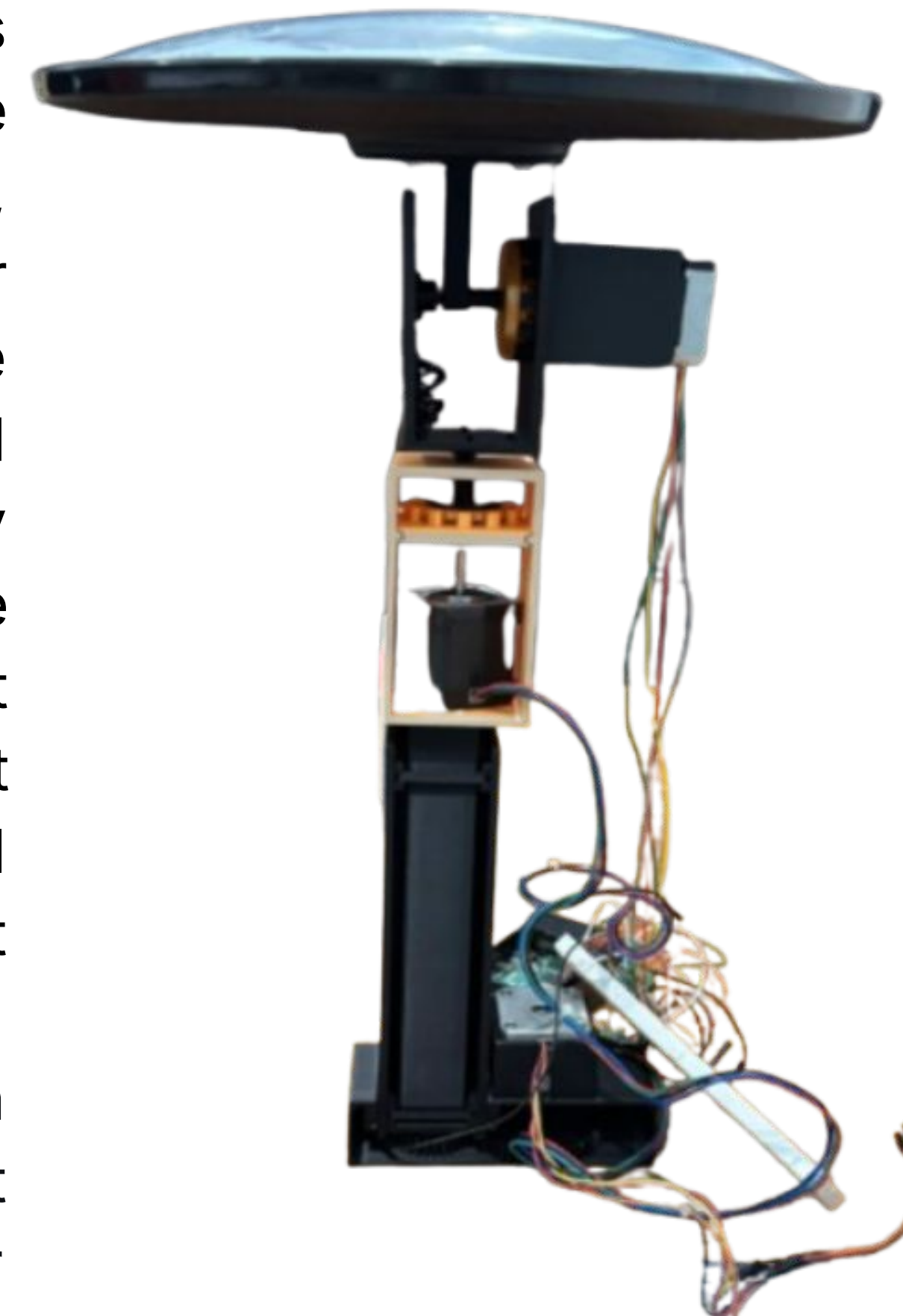


Fig. 3: Completed Heliostat

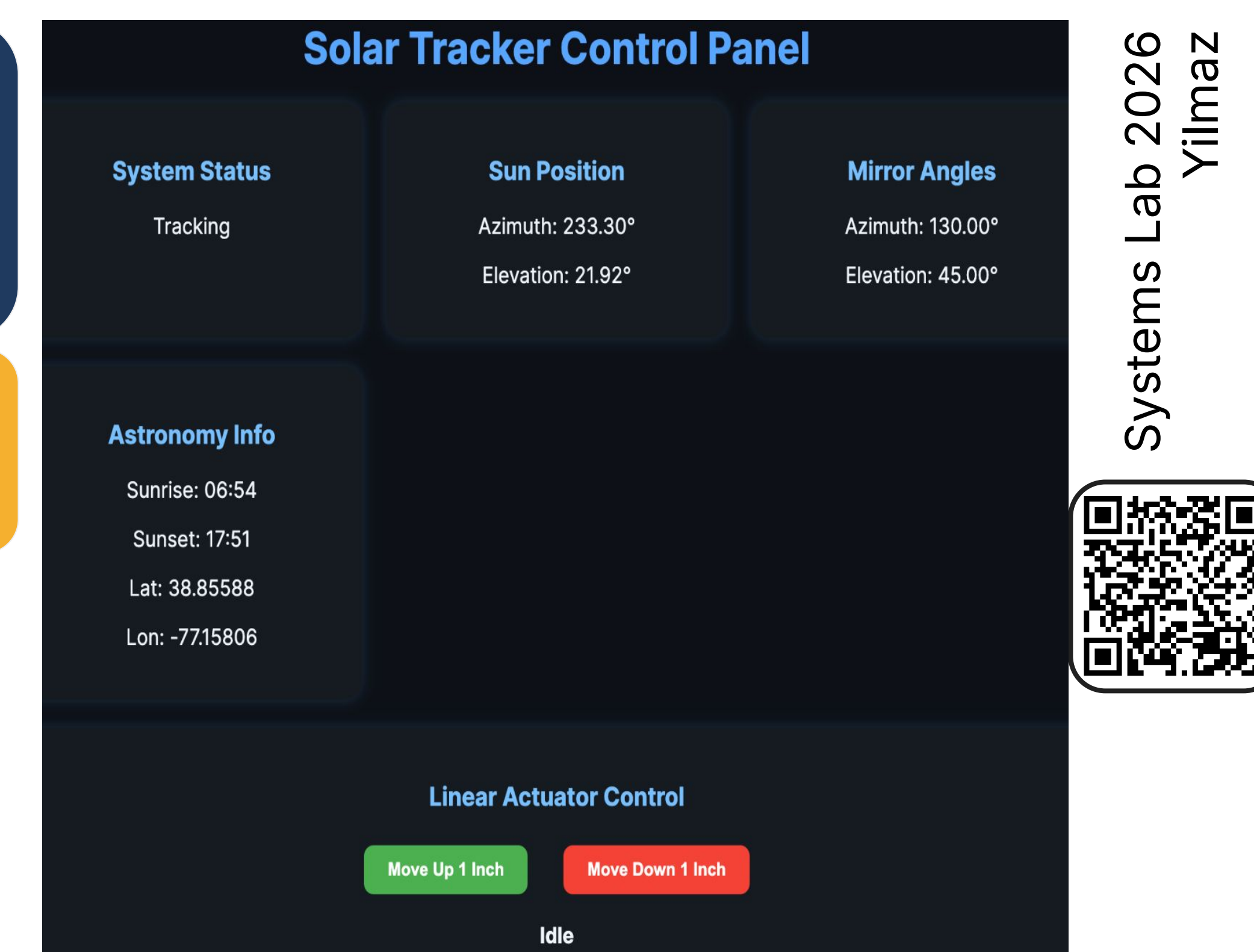


Fig. 4: User Interface

Results

The heliostat system uses a Raspberry Pi 4 to control three stepper motors for height, pitch, and yaw adjustment. The usage of the 1:11 cycloidal drive gear ratio made it possible for systems with high torque demands to rotate with small stepper motors. Motor drivers, external power delivery, and Wi-Fi solar position data were integrated to make automated multi-axis solar tracking possible.

Impact & Future Work

This compact, multi-axis micro-heliostat tracks the sun, improving light redirection throughout the day compared to stationary garden mirrors. It has the potential to increase solar panel efficiency, extend sunlight exposure for plants, and reduce reliance on artificial indoor lighting during the day.

Some future improvements include upgrading from plastic to metal components for improved durability, integrating an iris system to protect the mirror from dust and nighttime conditions, and developing a method in the user interface that provides optimized placement recommendations for maximum performance.